

ASHUTOSH BHARDWAJ

2A/148 Avas Vikas, Bulandshahr, U.P 203001

☎ 9536042121

✉ ashutosh24135@iiitd.ac.in

🌐 [linkedin.com/in/ashutosh-bhardwaj](https://www.linkedin.com/in/ashutosh-bhardwaj)

🐙 github.com/astral-fi

Education

Indraprastha Institute of Information Technology (IIIT Delhi)

New Delhi, India

Bachelor of Technology in Computer Science Engineering (CGPA: 8.7/10.0)

Aug. 2024 – Present

Experience

Undergraduate Research Intern

June 2025 – Present

Advisor: Dr. Sayan Basu Roy

New Delhi, Delhi

- Architected an end-to-end simulation-to-real (Sim2Real) training pipeline using ROS Noetic and Gazebo to synthesize 10k+ drone navigation data points in diverse virtual environments.
- Trained and evaluated deep neural control policies for autonomous navigation, achieving a 15% improvement in trajectory tracking accuracy over CNN and LSTM baselines.
- Deployed validated models onto an NVIDIA Jetson Nano-powered agricultural drone, facilitating autonomous on-board flight testing and real-world edge computing validation.

Undergraduate Research Intern

May 2026 – Present

Advisor: Dr. Saket Anand and Dr. Ojaswa Sharma

New Delhi, Delhi

- Building an Extended Reality (XR) testbed for mobile robotics by creating a static 3D digital twin of an indoor laboratory environment using 3D Gaussian Splatting (3DGS).
- Integrating multi-camera sensing pipelines for real-time robot localization and pose estimation within the reconstructed environment
- Evaluating system robustness through latency, fault-tolerance, and performance testing under varying sensing and communication conditions.

Projects

University ERP System | *Java, Swing, SQL, JDBC, OOP*

December 2025

- Engineered a high-concurrency Desktop ERP system utilizing Java Swing and JDBC, managing a dual-database architecture for authentication and academic record integrity.
- Developed a secure Role-Based Access Control (RBAC) system with SHA-256 password hashing to protect sensitive student data across 3+ administrative levels.
- Implemented automated modules for course registration and grade management, supporting over 100+ concurrent course entries and dynamic PDF transcript generation.

Visual Inertial Teach and Repeat | *Edge AI, Computer Visio*

April 2026

- Built a lightweight vision-based teach-and-repeat navigation system on Jetson Nano, integrating monocular perception, visual localization, and Ackermann Pure-Pursuit control for autonomous route following
- Designed an ORB + Lucas-Kanade tracking module for visual relocalization and heading correction, investigating trade-offs between feature-based and appearance-based navigation under low-texture and dynamic-scene conditions.
- Conducted experimental evaluation across multiple trajectories, analyzing drift, lateral error, route completion robustness, and resource utilization on embedded hardware.

SevaSetu: Government Navigator AI | *FastAPI, Docker, RAG, Pinecone*

August 2025

- Orchestrated a Retrieval-Augmented Generation (RAG) system using Scrapy and transformer-based embeddings to provide Hinglish-language support for 200+ government schemes.
- Developed a high-performance vector retrieval pipeline with Pinecone and FastAPI, improving query relevance and response accuracy for 1,000+ potential daily users.
- Containerized the full-stack application using Docker and deployed via Render and Twilio for seamless WhatsApp integration and cross-platform accessibility.

Technical Skills

Core Competencies: Computer Vision, Operating Systems, Computer Architecture and Organization, Data Structures and Algorithms, Object Oriented Programming, and Analysis and Design of Algorithms

Languages: Python, C++, C, Verilog, SQL (PostgreSQL/MySQL), Java, Bash

Tools & Frameworks: Git, Docker, ROS2, Gazebo, TensorFlow, PyTorch, FastAPI, NVIDIA Jetson SDK, FPGA, Vivado, Scrapy, Pinecone

Developer Tools: Linux (Ubuntu), VS Code, IntelliJ IDEA, Jupyter Notebook, Conda, Google Colab